APPLICATIONS OF

DEEP LEARNING

Agriculture

Blue River Technology
Blue River Technology has developed a robot called See & Spray which uses computer vision technologies like object detection to monitor and precisely spray weedicide on cotton plants. Precision spraying can help prevent herbicide resistance.

Plantix
Berlin-based agricultural tech start-up called PEAT, has developed an application called Plantix that identifies potential defects and nutrient deficiencies in the soil through images.

The image recognition app identifies possible defects through images captured by the user's smartphone camera. Users are then provided with soil restoration techniques, tips, and other possible solutions. The company claims that its software can achieve pattern detection with an estimated accuracy of up to 95%.



Self-Driving Cars

Tesla was one of the first automotive brands to launch a self-driving vehicle, and Audi, Cadillac, and Volvo are already developing their own models.

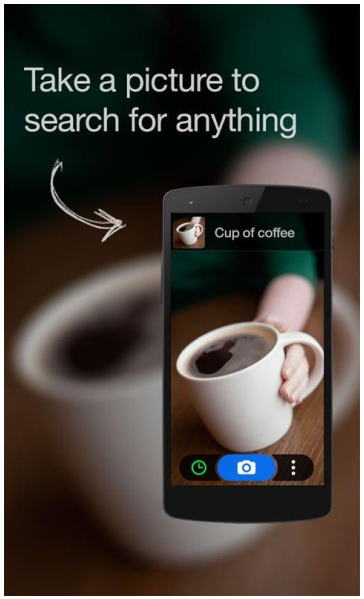
Uber's Self-Driving Truck Makes Its First Delivery
https://www.youtube.com/watch?v=sIlCR4eG8_o



Photographs That Become Purchases

CamFind - Visual Search Engine

Amazon, the giant e-commerce platform, has already incorporated a visual search option on its mobile application. Just take a photo of the item you want, and it will show you something very similar or identical.

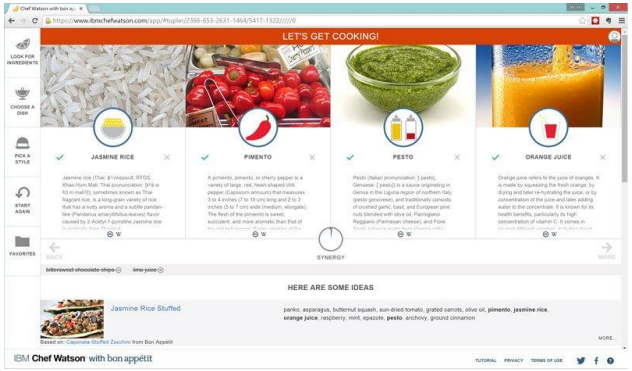


The Chef with The Most Recipes In the World

"Chef Watson's" artificial intelligence will help you create a recipe based on the ingredients you already have. Its algorithms will go through which dishes can be prepared with your ingredients and then give you several options so you can choose the dish you want, along with instructions on how to prepare it.

Chef Watson's creators fed it with nearly 10,000 recipes from *Bon Appetit's* archives and it used natural language processing to analyze the recipes to learn the underlying logic, understand the ingredients that are most commonly used and how ingredients were combined.
As it continues to process recipe requests, Chef Watson refines its algorithms by incorporating user feedback.

The professional version of Chef Watson takes it a step further. Researchers scraped the internet and added 30,000-plus recipes and added information about the molecular makeup of smells and flavour compounds in food.



The Best Flight

Hopper application

Through predictive analytics driven by artificial intelligence, the app is able to predict price patterns and alert travelers of the cheapest times to buy flights to their destinations. It monitors billions of prices every day and, based on the data history of each route, it can anticipate how the price trend will evolve. This way, you can configure notifications that'll remind you to book your flight at the exact time the price is right!

Health Care

An organization called Cambio Health Care developed a clinical decision support system for stroke prevention that can give the physician a warning when there's a patient at risk of having a heart stroke.

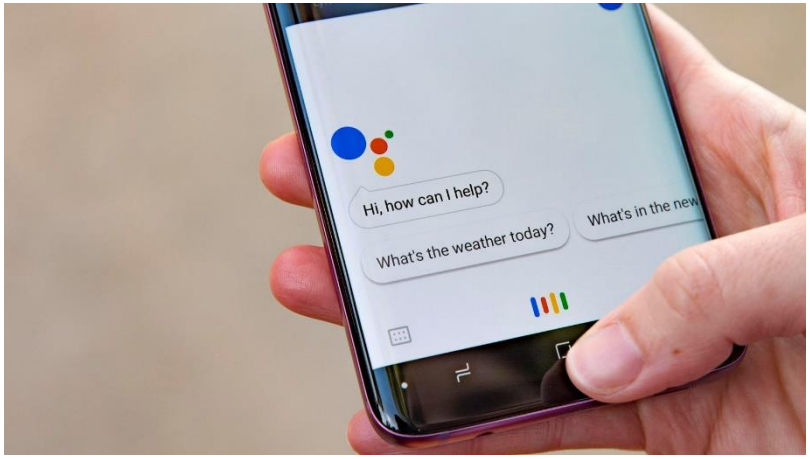


Chatbots

These days Virtual assistants have become a very common technology. A few examples include Siri, Cortana, Google's virtual assistant called Google Duplex.

Amazon's Echo is an example of how Artificial Intelligence can be used to translate human language into desirable actions. This device uses speech recognition and NLP to perform a wide range of tasks on your command. It can be used to control the devices at your house, book cabs, make phone calls, order your favorite food, check the weather conditions and so on.





Social Media

In social media platforms like Facebook, AI is used for face verification wherein machine learning and deep learning concepts are used to detect facial features and tag your friends.

Another such example is Twitter's AI, which is being used to identify hate speech and terroristic language in tweets. It makes use of Machine Learning, Deep Learning, and Natural language processing to filter out offensive content. The company discovered and banned 300,000 terrorist-linked accounts, 95% of which were found by non-human, artificially intelligent machines.